**CONFIDENTIAL INFORMATION**

# PFIZER

PSSM PCoE
VALIDATION REPORT
R&D MICROBIOLOGY - CHENNAI

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| Report Title: | Summary Report for Validation and Transfer of Microbial Enumeration Tests of Sugammadex Sodium to Rocky Mount, NC |
| Study No.: | VAL100190816 |
| Version: | 1.0 |
| Project Name/Number: | Sugammadex Sodium |

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## 1.0 PURPOSE

The purpose of this summary report is to summarize the results obtained from the Microbial Enumeration Tests for Sugammadex Sodium drug substance Manufactured by TEVA executed as per the Protocol No., VAL100188121 Version 1.0

## 2.0 SUMMARY

BRIDION (Sugammadex injection) is used for the reversal of neuromuscular blockade induced by rocuronium bromide and vecuronium bromide in adults undergoing surgery.

Sugammadex Sodium is the active ingredient in BRIDION (Sugammadex Injection). Pfizer identified TEVA as an API vendor to supply Sugammadex sodium to manufacture the Sugammadex Injection.

Sugammadex Sodium API monograph does not exist in the United States Pharmacopeia (USP), the European Pharmacopoeia (EP), the British Pharmacopoeia (BP) and the Japanese Pharmacopoeia (JP). However, as per Pfizer policy standard (V7112) Total Aerobic Microbial Count (TAMC) and Total Yeasts and Molds Count (TYMC) was included to have a process control of incoming materials bio load. The TAMC and TYMC testing were aimed at those microorganisms that can survive and/or proliferate in the drug product. The TAMC and TYMC method validation will establish an appropriate rinse scheme to adequately wash the drug substance solution from the membrane filters and allow the microbial recovery in the presence of drug substance.

Simultaneously, 3 batches Method validation and Qualification study was performed at Chennai Microbiology R&D (originating laboratory). The method validation was executed per the guidance of USP chapter <61> and <1227>.

## 3.0 MICROBIAL ENUMERATION TEST METHOD TRANSFER

The method validation was performed at Chennai R&D, the originating laboratory, on 3 different lots of Sugammadex Sodium Drug Substance manufactured by TEVA in triplicate by membrane filtration technique in accordance with validation protocol. The receiving laboratory is Rocky Mount Quality Control Microbiology.

The TAMC and TYMC method validation were performed using the membrane filtration method with 0.45 µm mixed cellulose ester membrane. This is simple and commonly used membrane filtration methodology: sample preparation was performed to the ratio of 1:10

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ratio of 10g of Sugammadex Sodium drug substance in 85 mL of sterile phosphate buffer solution pH 7.2 and added 5 mL of 0.1N HCl. Pre wetting of the membrane filter was performed with minimum of 25 mL Diluting Fluid A (DFA), post completion of pre-wetting the prepared sample was filtered and rinse scheme of 3x100 mL aliquot of the Diluting Fluid A (DFA) was carried out to adequately flush the drug substance residues from the filters.

Per SOP-LAB-03312 section 4.2.2, a formal method transfer study is not required for this method because the test method is very similar to the standard test method and materials that are already in use at the receiving laboratory. Therefore, the site method verification studies are not required, and the transfer is hereby waived.

## 4.0 MATERIALS

### 1) Media and Diluents:

| Name of media and Diluents | Make | Lot No. | Expiry Date / Use before date |
|---|---|---|---|
| Soybean Casein Digest Agar Plates | Hi-Media | MPB0656 | 31 JAN 2023 |
| Sabouraud Dextrose Agar Plates | Hi-Media | MPB0903 | 31 MAR 2023 |
| Phosphate Buffer pH 7.2 | In House | 00715609-0062A | 22 OCT 2022 |
| Diluting Fluid A (0.1% Peptone) | Hi-Media | 00715590-0026 A | NA* |
| Rehydration Fluid | Biomerieux | JDP-057 | 17 APR 2023 |
| 0.1N HCL | Merck | HC01429260 | 31 JAN 2023 |

* Diluting Fluid A sterilized using 0.22 µm filter during the validation study and used on the day of validation study.

### 2) Consumables:

| Consumables | Make | Lot No. | Expiry Date / Use before date |
|---|---|---|---|
| Sterile 0.45µm X 47 mm mixed cellulose ester membrane filters | Merck | F1EB16037 | 31 MAY 2023 |
| 0.22 µm filtration system (Stericup® 1000 mL Durapore® 0.22 µm) | Merck | MP211505G2 | 30 APR 2024 |
| Sterile Container (100mL) | Tarsons | 034BVH291020 | 28 OCT 2025 |
| Sterile micropipette Tips (1-10mL) | Eppendorff | H175855H | 28 FEB 2023 |
| Sterile micropipette tips 2-200 µL | AST | 21K22S208-302 | 28 FEB 2025 |
| Syringe filter 0.22 micron | Merck | R0JB89897 | 31 JUL 2023 |
| Sterile 10mL syringe | Dispovan | 225102JF2 | 31 MAY 2027 |

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### 3) Equipment's used:

| S. No | Name of Equipment | Equipment No. | Calibration due date | Performance Verification due date |
|---|---|---|---|---|
| 1 | Incubators | PCE1088 (22.5 ± 2.5°C) | 24 FEB 2023 | 25 FEB 2023 |
| | | PCE1087 (32.5 ± 2.5°C) | 24 FEB 2023 | 25 FEB 2023 |
| 2 | Micropipette | PCE0837 (1-10 mL) | 16 FEB 2023 | NA |
| | | PCE0835 (10-100 µL) | 10 MAR 2023 | NA |
| 3 | Vortex mixer | PCE0882 | NA | NA |
| 4 | Analytical balance | PCE1057 | 13 APR 2023 | NA |
| 5 | pH meter | PCE1059 | 16 MAR 2023 | NA |
| 6 | Biosafety cabinet | PCE0811 | NA | 05 NOV 2022 |
| 7 | Colony counter | PCE0846 | 11 MAR 2023 | NA |
| | | PCE0848 | 11 MAR 2023 | |
| 8 | EZ-Stream Pump | PCE0887 | NA | NA |
| 9 | Master flex Digital Pump Drive | PCE0673 | NA | NA |

### 4) Microorganisms Used:

**Table-1**

| Name of Microorganism | Make | Lot Number | Expiry Date |
|---|---|---|---|
| *Bacillus subtilis* NCTC 10400 (ATCC 6633) | Biomerieux | B 6365 | 02 NOV 2022 |
| *Pseudomonas aeruginosa* NCTC 12924 (ATCC 9027) | Biomerieux | B 6456 | 17 DEC 2022 |
| *Staphylococcus aureus* NCTC 10788 (ATCC 6538) | Biomerieux | B 6502 | 20 JAN 2023 |
| *Candida albicans* NCPF 3179 (ATCC 10231) | Biomerieux | B 6397 | 19 NOV 2022 |
| *Aspergillus brasiliensis* NCPF 2275 (ATCC 16404) | Biomerieux | B 6412 | 26 NOV 2022 |

## 5.0 METHODS/PROCEDURES

### A. Microbial Enumeration Test:

Per protocol simultaneously, 3 batches method validation and qualification study were executed in triplicates and the results were summarized in this report

**Sample Quantity: 10g.**

The following sample was taken for validation and Qualification.

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| Drug substance | Supplier | Batch/lot number |
|---|---|---|
| Sugammadex Sodium | TEVA | 7MPS02844051221 |
| | | 7MPS02844061221 |
| | | 7MPS02845860319 |

**Sample preparation for Microbial Enumeration Test:**

Samples of the Sugammadex Sodium drug substances were weighed in separate sterile containers as follows:

| S. No. | Batch Number of the sample | Weight of sample taken | | |
|---|---|---|---|---|
| | | Container 1 | Container 2 | Container 3 |
| 1 | 7MPS02844051221 | 10.02519 g | 10.09143 g | 10.02735 g |
| 2 | 7MPS02844061221 | 10.06623 g | 10.05833 g | 10.00739 g |
| 3 | 7MPS02845860319 | 10.01126 g | 10.03348 g | 10.02778 g |

85 mL of sterile Phosphate buffer pH 7.2 and 5mL of 0.1N HCL was added to each weighed sample containers of 3 different Batch of sugammadex Sodium drug substance (Batch Number: 7MPS02844051221, Batch Number: 7MPS02844061221 and Batch Number: 7MPS02845860319). Each sample container was mixed by vortex mixer until the drug substance get dissolved and filtered the test solution with 0.22 µm filtration system. With this ratio, 10mL of test solution contains approximately 1g of the drug substance.

This 1:10 dilution was termed as Solution-S

**Note: Sample solution (1:10) used for Product Negative controls were not sterile filtered as to capture inherit bioburden of the raw material.**

3 different Batch number of Sugammadex Sodium Solution-S (10mL) was dispensed separately, and pH was checked, the observed results were within the acceptance criteria of 6-8.

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**Observed pH values as follows:**

| S. No. | Batch Number of the sample | pH of the sample |
|---|---|---|
| 1 | 7MPS02844051221 | 7.333 |
| 2 | 7MPS02844061221 | 7.336 |
| 3 | 7MPS02845860319 | 7.335 |

## 5.1. PRODUCT POSITIVE CONTROL

5.1.1. Three filter membranes were pre-wet with 25 mL of sterile Diluting Fluid A and contents were filtered.

5.1.2. While vacuum pump in off condition, 10 ml of sample Solution-S was added to each filter membrane and 0.1 ml of microbial suspension (*B.subtilis*) of NMT 100 CFU / 0.1 mL was inoculated in sample solution. Filtered the contents.

5.1.3. Each filter membrane was rinsed with 3 X 100 mL quantity of sterile Diluting Fluid A.

5.1.4. The steps 5.1.1 to 5.1.3 were repeated for the remaining microorganisms as given in the Table-1.

5.1.5. Each filter membrane containing bacterial, yeast and mold cultures was aseptically transferred on to SCDA plate and incubated at 30-35˚C for 2 days. Similarly, the steps 5.1.1 to 5.1.3 were repeated for yeast & mold and each filter membrane was aseptically transferred on to SDA plates and incubated at 20-25˚C for 3 days.

5.1.6. The steps 5.1.1 to 5.1.5 were repeated for the other two batches of Sugmmadex Sodium drug substance.

## 5.2. PRODUCT NEGATIVE CONTROL

5.2.1. Three filter membranes were pre-wet with 25 mL of sterile Diluting Fluid A and contents were filtered**.**

5.2.2. 10 mL aliquot of Solution-S (unfiltered) was transferred to each of three filter membranes and filtered the solution.

5.2.3. Each filter membrane was rinsed with 3 X 100 mL quantity of sterile Diluting Fluid A.

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5.2.4. Each filter membrane was aseptically transferred on to SCDA plate and incubated at 30-35˚C for NLT 5 days. Similarly, the steps 5.2.1 to 5.2.3 were repeated, and the membrane was aseptically transferred on to SDA plates and incubated at 20-25˚C for NLT 7 days.

5.2.5. The steps 5.2.1 to 5.2.4 were repeated for the other two batches of Sugmmadex Sodium drug substance**.**

## 5.3. BUFFER NEGATIVE CONTROL

5.3.1. Three filter membranes were pre-wet with 25 mL of sterile Diluting Fluid A and contents were filtered**.**

5.3.2. 10 mL aliquot of Phosphate Buffer pH 7.2 was transferred to each of three filter membranes and filtered the solution. Each filter membrane was rinsed with 3 X 100 mL quantity of sterile Diluting Fluid A.

5.3.3. Each filter membrane was aseptically transferred on to SCDA plate and incubated at 30-35˚C for NLT 5 days. Similarly, repeated the step 5.3.1 to 5.3.2 and each filter membrane was aseptically transferred on to SDA plates and incubated at 20-25˚C for NLT 7 days**.**

## 5.4. PEPTONE NEGATIVE CONTROL

5.4.1. Three filter membranes were pre-wet with 25 mL of sterile Diluting Fluid A and contents were filtered.

5.4.2. 10 mL aliquot of Diluting Fluid A was transferred to each of three filter membranes and filtered the solution. Each filter membrane was rinsed with 3 X 100 mL quantity of sterile Diluting Fluid A.

5.4.3. Each filter membrane was aseptically transferred on to SCDA plate and incubated at 30-35˚C for NLT 5 days. Similarly, the steps 5.4.1 to 5.4.2 were repeated, and each filter membrane was aseptically transferred on to SDA plates and incubated at 20-25˚C for NLT 7 days**.**

## 5.5. INOCULUM COUNT VERIFICATION

5.5.1. Three filter membranes were pre-wet with 25 mL of sterile Diluting Fluid A and contents were filtered.

5.5.2. While vacuum pump in off condition, 100 mL of sterile Diluting Fluid A was added to each filter and NMT 100 cfu/ 0.1 mL of microbial suspension (*B.subtilis*) was inoculated to each filter and filtered the solution.

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5.5.3. The steps 5.5.1 to 5.5.2 were repeated for the remaining four organisms given in the Table-1.

5.5.4. Each of three filter membrane containing bacterial, yeast and mold cultures was aseptically transferred on to SCDA plate and incubated at 30-35˚C for 4 days (interim count taken to ensure countable colonies). Similarly the steps 5.5.1 to 5.5.3 were repeated again for yeast & mold and each filter membrane was aseptically transferred on to SDA plates and incubated at 20-25˚C for 7 days**.**

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## 5.6. RESULTS

### 5.6.1. Results for Batch number: 7MPS02844051221 (Batch-1)

- **Results observed on SCDA and SDA plates:**

| Date of Testing: 03 OCT 2022 | Test & Media Name: | PPC-SCDA | IC-SCDA | PPC-SDA | IC-SDA |
|---|---|---|---|---|---|
| | Incubation Completed on: | 05 OCT 2022 | 07 OCT 2022 | 06 OCT 2022 | 10 OCT 2022 |

| Results observed on SCDA Plates | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Microorganisms | Count observed on Product Positive Control (PPC) in CFU | | | | Count observed on Inoculum Count (IC) in CFU | | | | % Recovery (Average PPC/Average IC) x 100 |
| | Plate 1 | Plate 2 | Plate 3 | Average | Plate 1 | Plate 2 | Plate 3 | Average | |
| *Bacillus subtilis* NCTC 10400 (ATCC 6633) | 44 | 36 | 35 | **38.33** | 39 | 35 | 34 | **36.00** | **106%** |
| *Pseudomonas aeruginosa* NCTC 12924 (ATCC 9027) | 41 | 39 | 42 | **40.67** | 40 | 38 | 43 | **40.33** | **101%** |
| *Staphylococcus aureus* NCTC 10788 (ATCC 6538) | 46 | 42 | 42 | **43.33** | 46 | 43 | 42 | **43.67** | **99%** |
| *Candida albicans* NCPF 3179 (ATCC 10231) | 44 | 47 | 42 | **44.33** | 46 | 44 | 43 | **44.33** | **100%** |
| *Aspergillus brasiliensis* NCPF 2275 (ATCC 16404) | 37 | 35 | 36 | **36.00** | 36 | 38 | 35 | **36.33** | **99%** |
| **Results observed on SDA Plates** | | | | | | | | | |
| *Candida albicans* NCPF 3179 (ATCC 10231) | 44 | 45 | 46 | **45.00** | 50 | 56 | 54 | **53.33** | **84%** |
| *Aspergillus brasiliensis* NCPF 2275 (ATCC 16404) | 35 | 34 | 33 | **34.00** | 40 | 39 | 36 | **38.33** | **89%** |

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5.6.2. **Results for Batch number: 7MPS02844061221 (Batch-2)**

➢ **Results observed on SCDA and SDA plates:**

| Date of Testing: 03 OCT 2022 | Test & Media Name: | PPC-SCDA | IC-SCDA | PPC-SDA | IC-SDA |
|---|---|---|---|---|---|
| | Incubation Completed on: | 05 OCT 2022 | 07 OCT 2022 | 06 OCT 2022 | 10 OCT 2022 |

| Results observed on SCDA Plates | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Microorganisms | Count observed on Product Positive Control (PPC) in CFU | | | | Count observed on Inoculum Count (IC) in CFU | | | | % Recovery (Average PPC/Average IC) x 100 |
| | Plate 1 | Plate 2 | Plate 3 | Average | Plate 1 | Plate 2 | Plate 3 | Average | |
| *Bacillus subtilis* NCTC 10400 (ATCC 6633) | 41 | 40 | 35 | **38.67** | 39 | 35 | 34 | **36.00** | **107%** |
| *Pseudomonas aeruginosa* NCTC 12924 (ATCC 9027) | 43 | 38 | 42 | **41.00** | 40 | 38 | 43 | **40.33** | **102%** |
| *Staphylococcus aureus* NCTC 10788 (ATCC 6538) | 47 | 40 | 45 | **44.00** | 46 | 43 | 42 | **43.67** | **101%** |
| *Candida albicans* NCPF 3179 (ATCC 10231) | 43 | 42 | 38 | **41.00** | 46 | 44 | 43 | **44.33** | **92%** |
| *Aspergillus brasiliensis* NCPF 2275 (ATCC 16404) | 32 | 29 | 30 | **30.33** | 36 | 38 | 35 | **36.33** | **83%** |
| **Results observed on SDA Plates** | | | | | | | | | |
| *Candida albicans* NCPF 3179 (ATCC 10231) | 47 | 45 | 41 | **44.33** | 50 | 56 | 54 | **53.33** | **83%** |
| *Aspergillus brasiliensis* NCPF 2275 (ATCC 16404) | 33 | 34 | 31 | **32.67** | 40 | 39 | 36 | **38.33** | **85%** |

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5.6.3. **Results for Batch number: 7MPS02845860319 (Batch-3)**

➢ **Results observed on SCDA and SDA plates:**

| Date of Testing: 03 OCT 2022 | Test & Media Name: | PPC-SCDA | IC-SCDA | PPC-SDA | IC-SDA |
|---|---|---|---|---|---|
| | Incubation Completed on: | 05 OCT 2022 | 07 OCT 2022 | 06 OCT 2022 | 10 OCT 2022 |

| Results observed on SCDA Plates | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Microorganisms | Count observed on Product Positive Control (PPC) in CFU | | | | Count observed on Inoculum Count (IC) in CFU | | | | % Recovery (Average PPC/Average IC) x 100 |
| | Plate 1 | Plate 2 | Plate 3 | Average | Plate 1 | Plate 2 | Plate 3 | Average | |
| *Bacillus subtilis* NCTC 10400 (ATCC 6633) | 36 | 39 | 31 | **35.33** | 39 | 35 | 34 | **36.00** | **98%** |
| *Pseudomonas aeruginosa* NCTC 12924 (ATCC 9027) | 39 | 38 | 36 | **37.67** | 40 | 38 | 43 | **40.33** | **93%** |
| *Staphylococcus aureus* NCTC 10788 (ATCC 6538) | 51 | 48 | 45 | **48.00** | 46 | 43 | 42 | **43.67** | **110%** |
| *Candida albicans* NCPF 3179 (ATCC 10231) | 39 | 40 | 40 | **39.67** | 46 | 44 | 43 | **44.33** | **89%** |
| *Aspergillus brasiliensis* NCPF 2275 (ATCC 16404) | 33 | 30 | 28 | **30.33** | 36 | 38 | 35 | **36.33** | **83%** |
| **Results observed on SDA Plates** | | | | | | | | | |
| *Candida albicans* NCPF 3179 (ATCC 10231) | 51 | 44 | 43 | **46.00** | 50 | 56 | 54 | **53.33** | **86%** |
| *Aspergillus brasiliensis* NCPF 2275 (ATCC 16404) | 32 | 30 | 31 | **31.00** | 40 | 39 | 36 | **38.33** | **81%** |

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➢ **Equivalency percentage Recovery Calculation for mold & yeast between SCDA of PPC plate vs SDA of IC Plate:**

| Batch Number: 7MPS02844051221 (Batch-1) | | | |
|---|---|---|---|
| **Name of the organisms** | **Average PPC in SCDA (Average Value)** | **Average IC in SDA (Average Value)** | **Percentage recovery= (Average PPC in SCDA /Average IC in SDA) *100** |
| *Candida albicans* | 44.33 | 53.33 | **83%** |
| *Aspergillus brasiliensis* | 36.00 | 38.33 | **94%** |

| Batch Number: 7MPS02844061221 (Batch-2) | | | |
|---|---|---|---|
| **Name of the organisms** | **Average PPC in SCDA (Average Value)** | **Average IC in SDA (Average Value)** | **Percentage recovery= (Average PPC in SCDA /Average IC in SDA) *100** |
| *Candida albicans* | 41.00 | 53.33 | **77%** |
| *Aspergillus brasiliensis* | 30.33 | 38.33 | **79%** |

| Batch Number: 7MPS02845860319 (Batch-3) | | | |
|---|---|---|---|
| **Name of the organisms** | **Average PPC in SCDA (Average Value)** | **Average IC in SDA (Average Value)** | **Percentage recovery= (Average PPC in SCDA /Average IC in SDA) * 100** |
| *Candida albicans* | 39.67 | 53.33 | **74%** |
| *Aspergillus brasiliensis* | 30.33 | 38.33 | **79%** |

➢ **Results of Product Negative Control on SCDA Plates:**

Date of Testing: 03 OCT 2022 Incubation completed (PNC): 10 OCT 2022

| Batch Number of the product | Plate 1 | Plate 2 | Plate 3 | Average |
|---|---|---|---|---|
| 7MPS02844051221 | 0 | 0 | 0 | 0 |
| 7MPS02844061221 | 0 | 0 | 0 | 0 |
| 7MPS02845860319 | 0 | 0 | 0 | 0 |

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➢ **Results of Product Negative Control on SDA Plates:**

Date of Testing: 03 OCT 2022 Incubation completed (PNC): 11 OCT 2022

| Batch Number of the product | Plate 1 | Plate 2 | Plate 3 | Average |
|---|---|---|---|---|
| 7MPS02844051221 | 0 | 0 | 0 | 0 |
| 7MPS02844061221 | 0 | 0 | 0 | 0 |
| 7MPS02845860319 | 0 | 0 | 0 | 0 |

➢ **Results of Buffer Negative Control and Peptone Negative Control on SCDA Plates:**

Date of Testing: 03 OCT 2022 Incubation completed : 10 OCT 2022

| Test name | Plate 1 | Plate 2 | Plate 3 |
|---|---|---|---|
| Buffer Negative Control | No Growth Observed | No Growth Observed | No Growth Observed |
| Peptone Negative Control | No Growth Observed | No Growth Observed | No Growth Observed |

➢ **Results of Buffer Negative Control and Peptone Negative Control on SDA Plates:**

Date of Testing: 03 OCT 2022 Incubation completed : 11 OCT 2022

| Test name | Plate 1 | Plate 2 | Plate 3 |
|---|---|---|---|
| Buffer Negative Control | No Growth Observed | No Growth Observed | No Growth Observed |
| Peptone Negative Control | No Growth Observed | No Growth Observed | No Growth Observed |

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## 6.0 VALIDITY CRITERIA

Based on the 3 batches method validation results, the following validity criteria were found satisfactory as per protocol**:**

| Validity criteria | Observed Results |
|---|---|
| The morphology of the growth observed in the Product Positive Control and Inoculum Count Verification plates shall be consistent with the challenge microorganism(s) | The morphology of the growth observed in the Product Positive Control and Inoculum Count Verification plates were comparable with the challenge microorganism(s) |
| The Peptone negative control, Buffer negative control must be negative for growth | All negative controls (Buffer negative control and Peptone negative control) showed no growth. |
| The Product negative control must be not more than (NMT) 100 CFU/g | 1. Batch-1 (7MPS02844051221):<br>• SCDA plate – No growth<br>• SDA plate – No growth<br>2. Batch-2 (7MPS02844061221):<br>• SCDA plate – No growth<br>• SDA plate – No growth<br>3. Batch-3 (7MPS02845860319):<br>• SCDA plate – No growth<br>• SDA plate – No growth<br>The observed Total Aerobic Count limit met the acceptance criteria of NMT 100CFU/g. |
| The Inoculum Count verification plate's averages must be NMT 100 CFU | The inoculum count verification plate's averages are NMT 100 CFU/0.1mL. |
| The volume of the inoculum does not exceed 1% of the total volume of diluted product | The volume of the inoculum has not exceeded 1% of the total volume of the diluted product. |
| The exposed Settle Plates during validation (Set-1, Set-2 and Set-3) showed No Growth | |

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## 7.0 ACCEPTANCE CRITERIA

Microbial Enumeration Test method validation was carried out by using 3 Batches in triplicate by membrane filtration technique and outcome as follows,

| Acceptance criteria (Validation) | Observed Results |
|---|---|
| The recovery of each microorganism in the inoculated product (Product positive control) must be ≥ 50% and ≤ 200% when compared to the inoculum count verification | The recovery of each microorganism in the inoculated product (Product positive control) was ≥ 50% and ≤ 200% when compared to the inoculum count verification results |
| For *Aspergillus brasiliensis* and *Candida albicans*, the Product Positive Controls are plated on SCDA/TSA and SDA: and the Inoculum Checks are also plated on SCDA/TSA and SDA. SDA is the standard growth media for mold and yeast. If the comparison between PPC on SCDA/TSA and IC on SDA for *Aspergillus brasiliensis* and *Candida albicans* meet acceptance criteria of ≥ 50% and ≤ 200%, then the routine bioburden test can be performed using only SCDA/TSA media at incubation conditions of 30-35°C for not less than (NLT) 3 days. This comparison will validate equivalency between SCDA/TSA and SDA for growth of yeasts and molds | Equivalency recovery study between SCDA (TSA) and SDA for growth of yeasts and molds criteria met for *Aspergillus brasiliensis* and *Candida albicans*; therefore, the routine bio burden test can be performed using only SCDA (TSA) media at incubation conditions of 30-35°C for not less than (NLT) 3 days |

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## 8.0 CONCLUSION

3 batches of Method validation and Qualification met the validity criteria and acceptance criteria, hence the validated method can be used for routine analysis.

Equivalency recovery study between SCDA (TSA) and SDA for growth of yeasts and molds criteria met for *Aspergillus brasiliensis* and *Candida albicans*; therefore, the routine bio burden test can be performed using only SCDA (TSA) media at incubation conditions of 30-35°C for not less than (NLT) 3 days.

Based on the above results it is recommended that Sugammadex Sodium manufactured by TEVA is qualified and shall be routinely analyzed using membrane filtration technique as follows.

**Routine Test Procedure:**

1. Add 85 mL of sterile Phosphate Buffer Solution pH 7.2 to 10 g of Sugammadex Sodium.
2. Add 5 mL of 0.1N HCL to the preparation. Mix until dissolved. With this ratio, 10 mL of the test solution is containing approximately 1g of the drug substance.
3. Pre-wet the membrane filters with approximately 25 mL of Diluting Fluid A and filter the contents.
4. Transfer 10 mL aliquots of the prepared solution on to each membrane filter and filter the contents.
5. Rinse each membrane filter with 3 X 100 mL of Diluting Fluid A (Diluting Fluid A).
6. Transfer each membrane on to SCDA plates and incubate at 32.5 ± 2.5 °C for not less than 3 days.

## 9.0 NON-CONFORMANCE

No Non-Conformance was observed during execution of study.

## 10.0 REFERENCES

| eLab Notebook | Analyst |
|---|---|
| 00715580-0009 | Prakash S (UID:3005711 & NTID: SP33) |

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| Document Number | Document Title | Effective/Sign Off Date |
|---|---|---|
| USP <61> | Microbiological Examination of Non -Sterile Products | 01 MAY 2021 |
| USP <1117> | Microbiological Best Laboratory Practices | 01 MAY 2021 |
| USP <1227> | Validation of Microbial Recovery from Pharmacopeial Articles | 01 MAY 2021 |
| SOP-LAB-03311 | Microbiological Test Requirements for Ingredients Used in the Manufacture of Parenteral Products Procedure, PEH R&D | 22 DEC 2020 |
| V7112 | Microbiological Methods Validation | 12 JUN 2020 |
| L6102 | Transfer of test method | 08 JUN 2018 |
| SOP-LAB-05213 | WRD Biovia Workbook-PCoE R&D | 09 OCT 2020 |
| SOP-LAB-04874 | Validation for Total Aerobic Microbial count (TAMC) and Total Yeast and Molds count (TYMC) for Active Pharmaceutical ingredients and Excipients – Lake forest and IKKT | 30 SEP 2022 |
| SOP-LAB-03312 | Biological Laboratory Test Method Validation Procedure, PEH R&D | 24 AUG 2022 |
| SOP-LAB-02515 | Generating Scientific Protocols and Reports in GDMS | 21 APR 2021 |
| VAL100188121 | Validation of Microbiological Examination of Non-Sterile Products: Microbial Enumeration Tests for Sugammadex Sodium API Manufactured by TEVA | 03 OCT 2022 |

## 11.0 CHANGE HISTORY

| Version | Description of Change(s) | Reason for Change(s) |
|---|---|---|
| 1.0 | NIL | First Issue |

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# Document Approval Record

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| Signed By: | Date(GMT) | Signing Capacity |
|---|---|---|
| S, Prakash | 26-Oct-2022 12:17:06 | Author Approval |
| Sacro, Takara | 28-Oct-2022 13:59:56 | Quality Control Review |
| K, SENTHIL KUMAR | 31-Oct-2022 04:58:16 | Manager Approval |
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